

- ### Lengths of Members ($\rightarrow$)

$$\bullet l_{CE} = l_{DE} = \sqrt{b^2 + \left(\frac{a}{2}\right)^2} \approx 0.3606 \text{ m} \quad \bullet l_{BC} = l_{AD} = 0.5 \text{ m}$$

$$A_y = -B_y \quad A_y = 6.67 \text{ kN}$$
$$F_{BE} = + (1.12) \left(\frac{0.4}{0.5} \right) = 8.90 \text{ kN}$$
$$A_x = 0 \text{ kN} \quad B_y = -6.67 \text{ kN}$$

Use of symmetry

$$F_{AD} = -F_{BC}$$

$$F_{DE} = -F_{CE} \quad F_{AE} = -F_{BE}$$

Internal

$$F_{CF} = 16.02 \text{ kN (C)}$$

$$F_{BC} = 11.12 \text{ kN (C)} \quad F_{AD} = 11.12 \text{ kN (T)}$$

$$F_{BE} = 8.90 \text{ kN (T)} \quad F_{DE} = 16.02 \text{ kN (T)}$$

$$F_{CD} = 0 \text{ kN} \quad F_{AE} = 8.90 \text{ kN (C)}$$

Finding the cross-sectional area of members

For A500 $S_y =$ many values but assume A500 steel is D-grade with $S_y = 250 \text{ MPa}$

$$F_{\max} = 16020 \text{ N} \quad SF = 3.5 \quad \sigma_{\text{allow}} = ? \quad A_{\min} = ? \quad W_{\text{truss}} = ?$$

$$\sigma_{\text{allow}} = \frac{S_y}{SF} = \frac{250}{3.5} = 71.4286 \text{ MPa}$$

$$\sigma_{\text{allow}} = \frac{F_{\max}}{A_{\min}} \therefore A_{\min} = \frac{F_{\max}}{\sigma_{\text{allow}}} = \frac{16020 \text{ N}}{71.4286 \text{ MPa}} = 224.28 \text{ mm}^2 = 224.28 \times 10^{-6} \text{ m}^2$$

$$\rho_{\text{steel}} \approx 7850 \text{ kg/m}^3$$

$$g = 9.81 \text{ m/s}^2$$

$$L_{\text{total}} = 2(0.5) + 2(0.3606) + 2(0.6) + 0.9$$

$$L_{\text{total}} = 3.3212 \text{ m}$$

$$W_{\text{truss}} = mg = \rho_{\text{steel}} V g = \rho_{\text{steel}} \cdot A_{\min} \cdot L_{\text{total}} \cdot g$$

$$W_{\text{truss}} = (7850)(224.28 \times 10^{-6})(3.3212)(9.81)$$

$$W_{\text{truss}} = 57.36 \text{ N}$$

$$A_{\min} = 224.28 \times 10^{-6} \text{ m}^2$$

$$W_{\text{truss}} \approx 57.36 \text{ N}$$

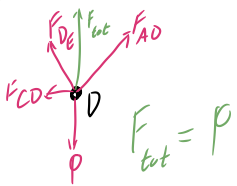
Finding the cross-sectional area of pins

$$\tau_y = 170 \text{ ksi} \quad SF = 4$$

$$\rho = 0.278 \text{ lb/in}^3$$

$$\tau_{\text{allow}} = ? \quad A_{\text{pin}} = ? \quad W_{\text{pin}} = ?$$

FBD



$$\tau_{\text{allow}} = \frac{\tau_y}{SF} = \frac{170}{4} = 42.5 \text{ ksi}$$

$$1 \text{ ksi} = 6.895 \text{ MPa}$$

$$42.5 \times 6.895 = 293.0375 \text{ MPa}$$

$$F_{\text{tot}} = 20 \text{ kN} = 20000 \text{ N}$$

$$\tau_{\text{allow}} = \frac{F_{\text{tot}}}{A_{\text{pin}}} \therefore A_{\text{pin}} = \frac{F_{\text{tot}}}{\tau_{\text{allow}}} = \frac{20000}{293.0375} = 68.25 \text{ mm}^2$$

$$1 \text{ meter} = 39.37 \text{ in}$$

$$1 \text{ m}^2 = 1549.9969 \text{ in}^2$$

for L use truss value of 0.015 m

$$W_{\text{pin}} = \rho \cdot A \cdot L$$

$$W_{\text{pin}} = 0.278 (0.106) (0.591) = (0.017) \text{ kips}$$

$$68.25 \times 10^{-6} \text{ m}^2 \times \frac{1549.9969}{1 \text{ m}^2} = 0.106$$

$$0.015 \text{ m} / \frac{39.37}{1 \text{ m}} = 0.591$$

$$A_{\text{pin}} = 68.25 \times 10^{-6} \text{ m}^2$$

$$W_{\text{pin}} = 0.087 \text{ lb} \cdot \text{in}$$

Determining ϕ & radius of pins & dimensions of truss system

$$A_{pin} = 68.25 \times 10^{-6} \text{ m}^2 \quad A_{truss} = 227.28 \times 10^{-6} \text{ m}^2$$

Assuming truss is square

Pin

$$A = \frac{\pi}{4} d^2$$
$$68.25 \times 10^{-6} = \frac{\pi}{4} d^2$$

$$\sqrt{\frac{4(68.25 \times 10^{-6})}{\pi}} = d_{pin}$$

$$9.32 \text{ mm}$$

or

$$0.0093 \text{ m}$$

truss

$$227.28 \times 10^{-6} = l^2 = A$$

$$l = \sqrt{227.28 \times 10^{-6}}$$
$$\approx 0.015 \text{ m}$$

$$\star 0.0093 + 0.015 = \phi \text{ of the joint that houses the pins} = 0.0243 \text{ m}$$